

VIP C2 PROGRAMME — FULL STACK DEVELOPMENT WITH
MERN

Data Flow Diagrams & User Stories

Online Complaint Registration and Management Systems — MERN
Stack E-commerce Application

Phase: Requirement Analysis Phase

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Prepared by: BHAVINI SURI

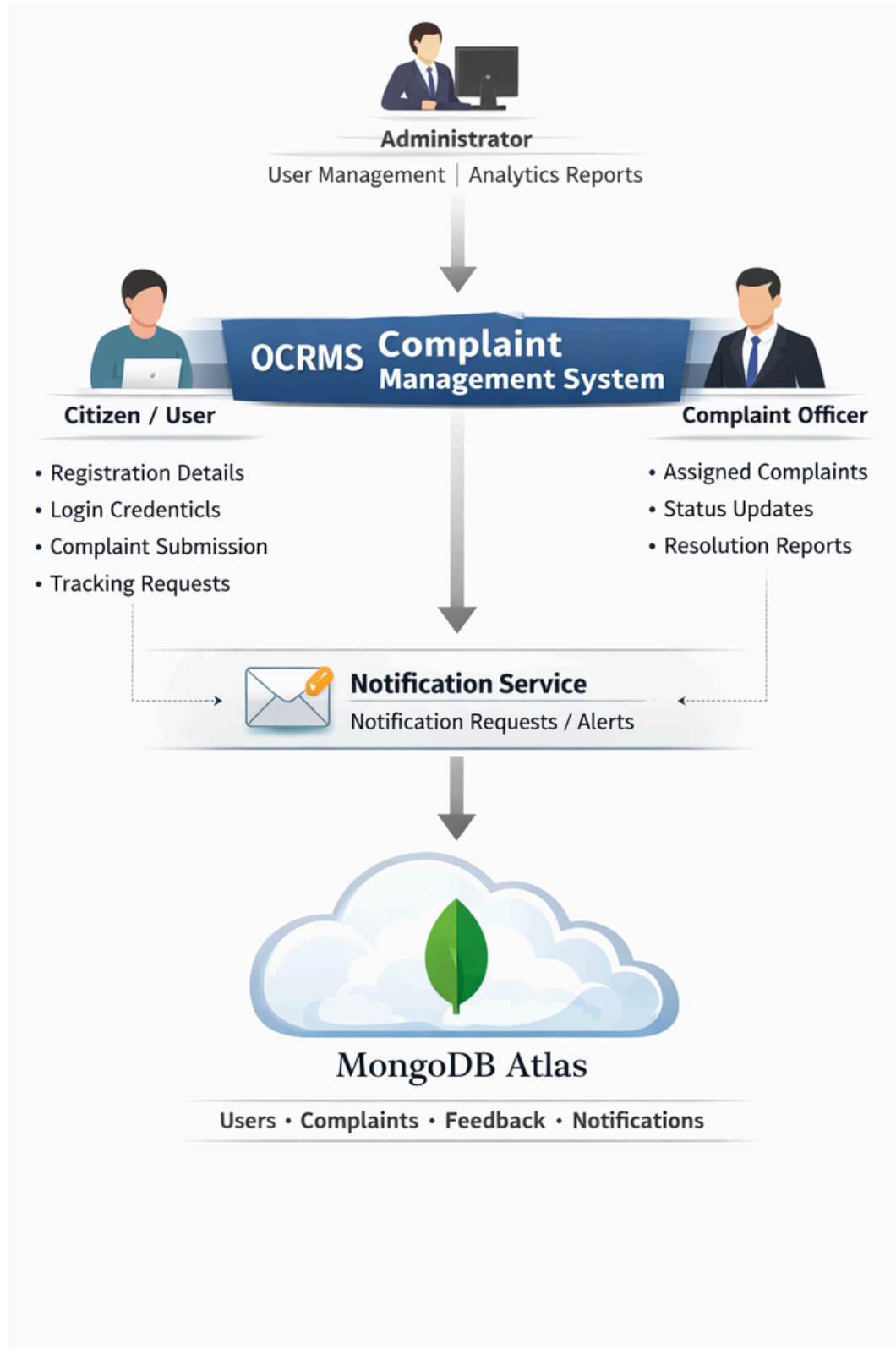
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Data Flow Diagrams & User Stories

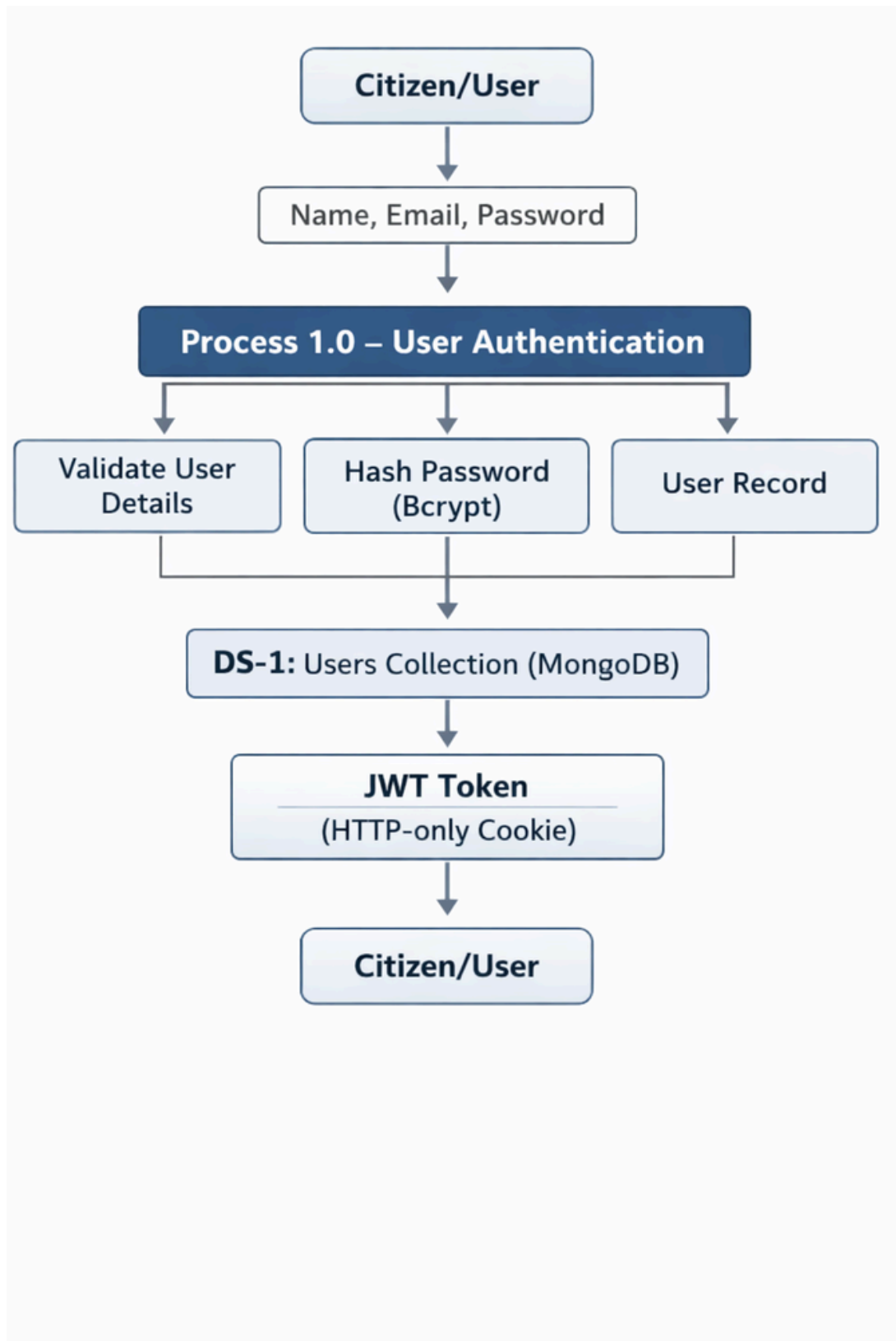
1. Introduction

This document presents the Data Flow Diagrams (DFDs) and formal User Stories for the Online Complaint Registration and Management System (OCRMS). DFDs model the system at two abstraction levels — Context Level (Level 0) and Functional Decomposition (Level 1) — illustrating data flows between users, administrators, complaint officers, processes, and data stores.

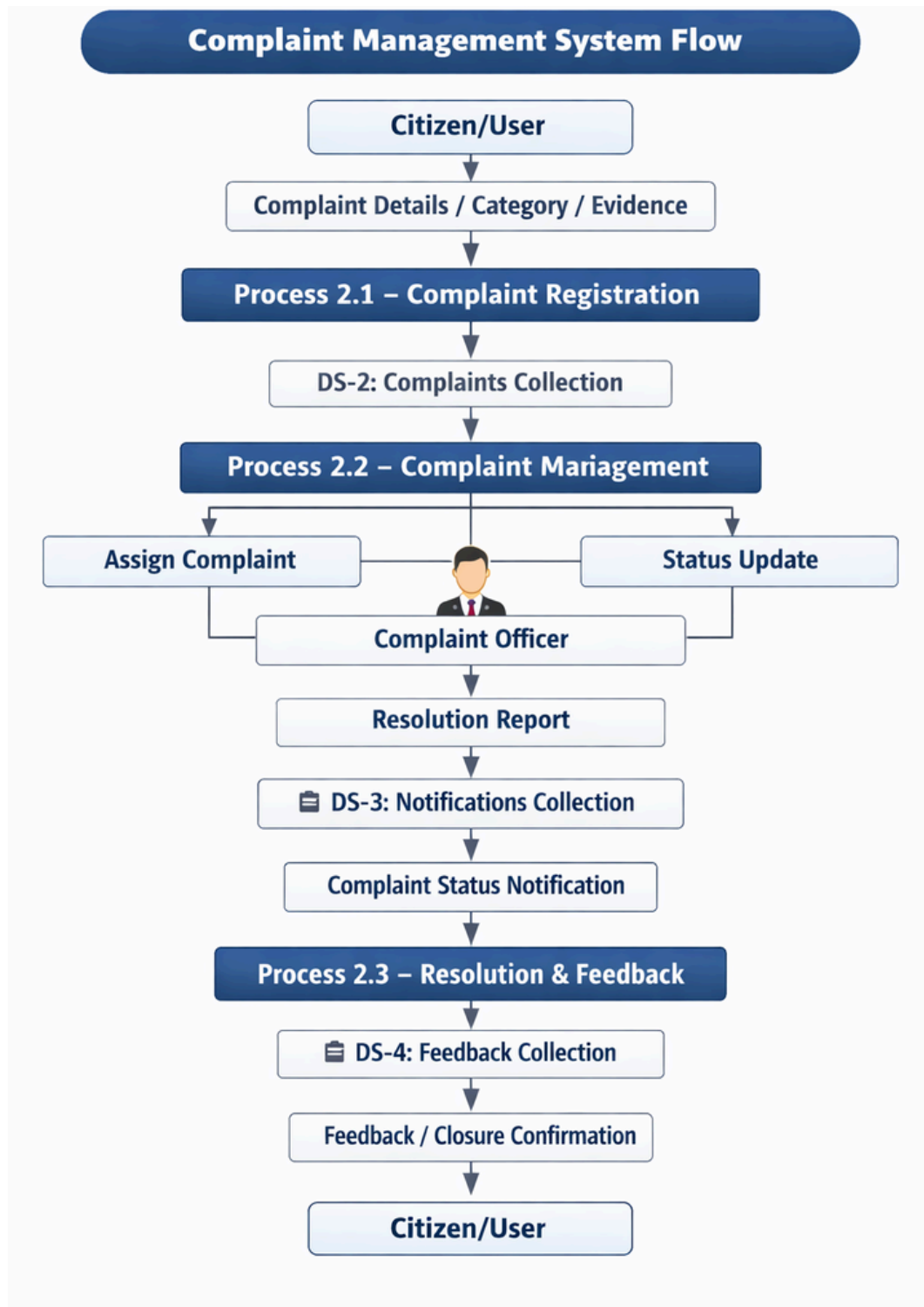
2. Level 0 – Context Diagram



3. Level 1 – Authentication Sub-Process



4. Level 1 – Complaint Registration And Resolution Sub- Process



5. User Stories

USN	User Type	User Story	Acceptance Criteria	Points	Sprint
USN-1	Citizen	As a citizen, I can register using my name, email, and password	Account created successfully; duplicate email rejected.	2	Sprint 1
USN-2	Citizen	As a citizen, I can log in securely using my credentials .	JWT generated; invalid credentials rejected.	2	Sprint 1
USN-3	Citizen	As a citizen, I can submit a complaint with category and description.	Complaint stored successfully and assigned ID generated	3	Sprint 1
USN-4	Citizen	As a citizen, I can track the status of my complaint.	Status updates displayed in real time.	3	Sprint 2

USN	User Type	User Story	Acceptance Criteria	Points	Sprint
USN-5	Citizen	As a citizen, I can communicate with assigned officers	Messages delivered successfully through chat system.	3	Sprint 2
USN-6	Complaint officer	As an officer, I can view assigned complaints and update status	Status changes reflected immediately to users	5	Sprint 2
USN-7	Citizen	As a citizen, I can submit feedback after complaint resolution	Feedback stored successfully and linked to complaint.	2	Sprint 3
USN-8	Administrator	As an administrator, I can manage users	User and officer records updated successfully	3	Sprint 1

USN	User Type	User Story	Acceptance Criteria	Points	Sprint
		and officers			
USN-9	Administrator	As an administrator, I can assign complaints to officers.	Assigned complaints appear in officer dashboard.	2	Sprint 3
USN-10	Administrator	As an administrator, I can view complaint analytics and reports.	Dashboard displays accurate complaint metrics and reports.	3	Sprint 3

6. Conclusion

The Data Flow Diagrams and User Stories provide a clear representation of how information moves through the OCRMS platform and how different users interact with the system. These artifacts help define functional requirements, system boundaries, and user expectations, ensuring efficient development and successful implementation of the complaint management solution